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MINISTRY OF EDUCATION & TRAINING
HOCHIMINH CITY UNIVERSITY OF TECHNOLOGY

FINAL ASSIGNMENT REPORT

[CMP179] - Software Testing and Quality Assurance

Lecturer's name: Mr. Trinh Cong Nhut

Student's name :

1. Trần Ngọc Ánh Dương Student's ID: 2280600534 Class: 22DTHQB2
2. Ao Thành Nhân Student's ID: 2280602150 Class: 22DTHQB2

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CHAPTER 1. TEST CAFÉ

1.1 Introduction:

TestCafe is a modern open-source end-to-end testing framework designed specifically for testing web applications. It allows developers and testers to automate user interactions with a website in order to verify whether the system behaves correctly under different scenarios.

Unlike many traditional automation tools, TestCafe is built on the Node.js platform and does not require external browser drivers such as WebDriver. This simplifies the installation process and reduces configuration complexity. Because of this architecture, TestCafe can run tests directly in the browser while managing communication between the test script and the browser internally.

TestCafe supports a wide range of modern browsers including:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

This cross-browser capability allows testers to verify whether a web application works consistently across different browser environments.

Another important feature of TestCafe is its automatic waiting mechanism. In many automation tools, testers must manually add waiting commands to ensure that page elements have loaded before interacting with them. TestCafe automatically waits for elements to appear and become interactable before performing actions such as clicking buttons or entering text. This significantly reduces synchronization errors in automated tests.

TestCafe test scripts are written in JavaScript or TypeScript, which makes the framework easy to integrate into modern web development environments. Since many web developers are already familiar with JavaScript, learning TestCafe is relatively straightforward.

Because of its simplicity, reliability, and strong browser automation capabilities, TestCafe has become a popular tool for automated UI testing in web applications.

1.2 TestCafe Project Structure:

A TestCafe project usually follows a clear and organized folder structure. Organizing the project properly helps developers maintain test scripts and manage dependencies efficiently.

The basic structure of a TestCafe project typically includes the following components:

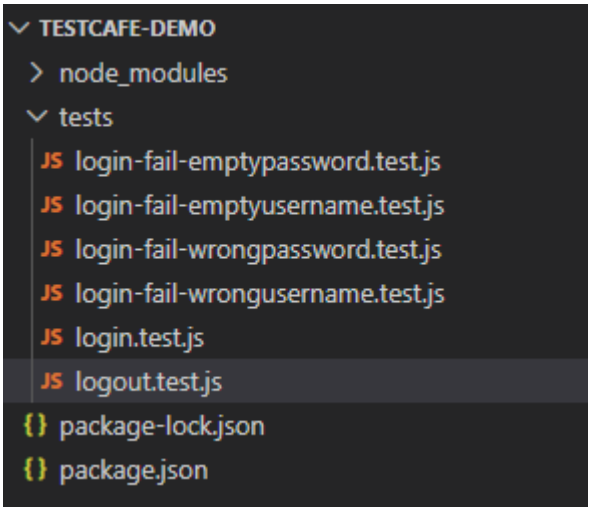

Figure 1.1 Basic structure of a TestCafe project

Each component plays an important role in the project.

1.2.1 package.json

The package.json file is automatically generated when initializing a Node.js project using npm.

This file has several important roles:

- Store project information such as name and version
- Manage libraries and dependencies used in the project
- Define scripts for running commands such as test execution

It acts as the central configuration file for the project.

1.2.2 node_modules

The node_modules folder contains all installed dependencies for the project, including TestCafe and other libraries.

This folder is automatically created when running the installation command.

1.3 Advantages and Limitations of TestCafe

1.3.1 Advantages

TestCafe provides several advantages that make it a useful tool for automated web testing.

1.3.1.1 Simple Installation

TestCafe is very easy to install compared with many traditional testing frameworks. Users only need to install Node.js and then run a simple npm command to install TestCafe.

There is no need to install browser drivers or configure additional tools.

1.3.1.2 Automatic Waiting Mechanism

TestCafe automatically waits for elements to appear on the page before interacting with them.

For example, when clicking a button or entering text into a field, TestCafe ensures that the element is visible and ready to receive interaction.

This greatly reduces common problems such as:

- Element not found errors
- Synchronization issues
- Timing failures

1.3.1.3 Cross-Browser Testing

TestCafe supports multiple browsers. The same test script can run on different browsers without modification.

This helps testers ensure that the web application works correctly across different environments.

1.3.1.4 Easy-to-Read Test Scripts

Because TestCafe tests are written in JavaScript, the syntax is easy to understand.

This makes the test scripts more readable and maintainable for both developers and testers.

1.3.1.5 Built-in Test Runner

TestCafe includes its own built-in test runner, which simplifies the process of executing tests and viewing results.

Users can run tests directly from the command line and immediately see whether the tests pass or fail.

1.3.2 Limitations

Despite its advantages, TestCafe also has some limitations.

1.3.2.1 Limited community compared with Selenium

Since Selenium has existed longer, it has a larger community and more resources.

1.3.2.2 Browser-based testing focus

TestCafe mainly focuses on web browser testing and may not be suitable for mobile native applications.

1.3.2.3 Performance limitations in large test suites

Very large projects may require additional configuration to manage test execution efficiently.

1.4 Comparison with Manual Testing

Criteria	Manual Testing	TestCafe Automation
Execution speed	Slow	Fast
Human effort	High	Low
Reusability	Limited	High
Accuracy	Can be affected by human error	Consistent and reliable
Test repetition	Time-consuming	Easily repeatable

Table 1.1 Comparison with Manual Testing

Manual testing is useful for exploratory testing and UI evaluation. However, automated testing tools like TestCafe are more effective for repetitive test cases such as login validation or form submission.

1.5 Simple Implementation

1.5.1 Project Setup

To use TestCafe, the computer must have Node.js installed, because TestCafe runs on the Node.js platform.

After preparing the environment, the first step is to create a project folder.

Next, initialize the project using the following command: npm init –y

This command automatically creates the package.json file.

The roles of the package.json file include:

- Storing project information
- Managing libraries and dependencies
- Supporting project-related commands

After this step, the project officially becomes a Node.js project.

Next, install TestCafe using the command: npm install testcafe --save-dev

TestCafe is installed as a dev dependency because it is only used for testing purposes and not for the production system.

After installation, the project will contain:

- the node_modules folder
- TestCafe listed inside package.json

To organize the code clearly, a folder named tests is created.

This folder stores all TestCafe test scripts.

Inside the tests folder, a file named login.test.js is created.

This file contains test cases for the login functionality.

1.5.2 Installing TestCafe

Next, TestCafe is installed using the following command: npm install testcafe --save-dev

TestCafe is installed as a development dependency because it is only required during the testing phase and is not used in the production system.

After installation, the project structure will include:

- node_modules folder
- package.json file containing the TestCafe dependency

1.5.3 Demonstration

In this demonstration, TestCafe is used to test the login functionality of the website: https://the-internet.herokuapp.com/login

The demonstration includes two test scenarios:

1. Failed Login Test
2. Successful Login Test

1.5.3.1 DEMO 1 – Failed Login Test

The first test demonstrates a failed login scenario, which is considered a negative test case.

The objective of this test is to verify whether the system correctly handles incorrect login attempts.

Test Steps:

1. Open the login page
2. Enter the correct username
3. Enter an incorrect password
4. Click the Login button

```
import { Selector } from 'testcafe';

fixture `Login Fail Demo`
    .page `https://the-internet.herokuapp.com/login`;

//wrong password
test('Login with wrong password', async t => {
    await t
        .typeText('#username', 'tomsmith')
        .wait(2000)
        .typeText('#password', 'wrongpassword')
        .wait(2000)
        .click('button[type="submit"]')
        .wait(2000)
        .expect(Selector('#flash').innerText)
        .contains('Your password is invalid!')
        .wait(9000)
});
```

Figure 1.2 TestCafe test script for failed login scenario using incorrect password

After performing these actions, TestCafe checks the error message displayed on the screen. The expected error message is: “Your password is invalid!”.

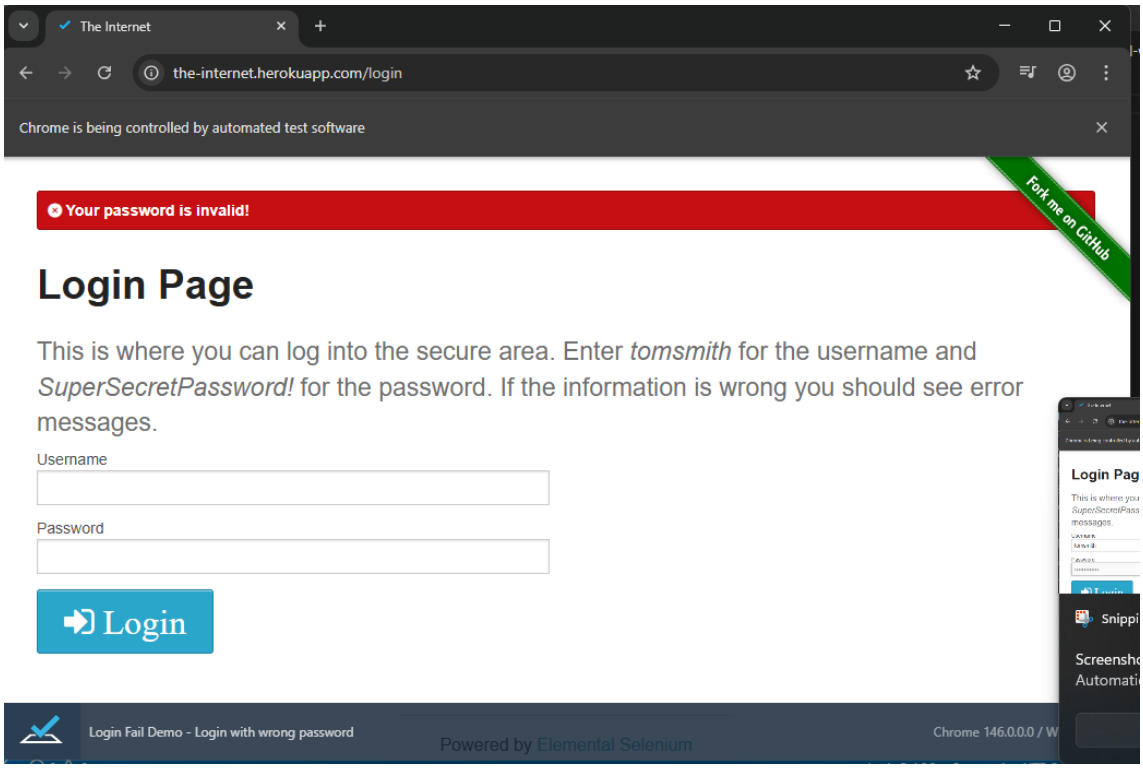

Figure 1.3 The expected error message

If the system displays this message, the test result is PASS. If the message does not appear or the content is incorrect, the test result is FAIL.

```
PS D:\Hutech\testcafe-demo\tests> npx testcafe chrome login-fail-wrongpassword.test.js
Running tests in:
- Chrome 146.0.0.0 / Windows 11

Login Fail Demo
√ Login with wrong password


1 passed (21s)
```

Figure 1.4 TestCafe execution result showing the successful completion of the "Login with wrong password" test case

This test confirms that the system correctly detects incorrect passwords and provides appropriate feedback to the user.

1.5.3.2 DEMO 2 – Successful Login Test

The second test demonstrates a successful login scenario, which is considered a positive test case.

Test Steps:

1. Open the login page
2. Enter the correct username
3. Enter the correct password
4. Click the Login button

```
import { Selector } from 'testcafe';

fixture `Demo Login Test`
    .page `https://the-internet.herokuapp.com/login`;

test('User login success', async t => {
    await t
        .typeText('#username', 'tomsmith')
        .wait(2000)
        .typeText('#password', 'SuperSecretPassword!')
        .wait(2000)
        .click('button[type="submit"]')
        .wait(4000)
        .expect(Selector('#flash').innerText)
        .contains('You logged into a secure area')
        .wait(9000)
});
```

Figure 1.5 TestCafe test script for successful login scenario

After a successful login, the system redirects the user to a secure page. The page displays the message: “You logged into a secure area”. TestCafe verifies that this message appears on the page.

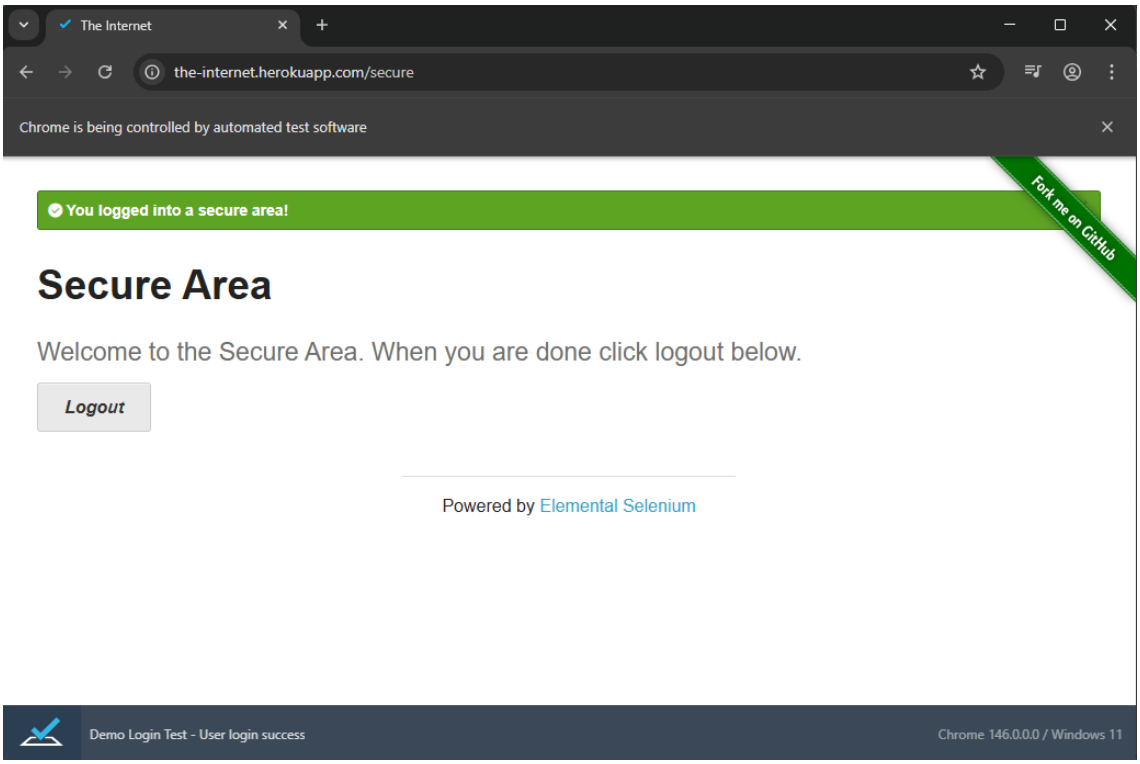

Figure 1.6 Successful login message displayed on the secure page

If the message is displayed correctly, the test case is marked as PASS, confirming that the login functionality works properly.

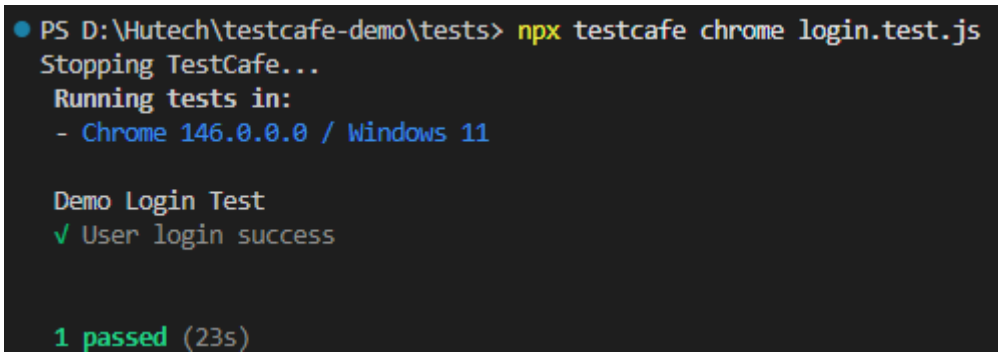

Figure 1.7 TestCafe execution result showing the successful completion of the “Successful Login” test case

This is a YouTube link introducing and demonstrating how to use the Appium application: https://youtu.be/t91zAR8lEgg?si=uPxjprKP3R-QZHmw

CHAPTER 2. SELENIUM IDE AUTOMATION TESTS

2.1 Introduction to Selenium IDE

Selenium IDE is a simple automation testing tool used for testing web applications directly in a web browser. It is provided as a browser extension and supports browsers such as Google Chrome and Microsoft Edge. Selenium IDE allows users to create automated tests without needing advanced programming knowledge.

One of the most important features of Selenium IDE is **the** record and playback capability. This feature allows testers to record their interactions with a website, such as clicking links, typing text into input fields, or navigating between pages. Selenium IDE automatically converts these actions into test commands that can later be replayed to verify whether the website functions correctly.

In Selenium IDE, automated tests are organized into test cases, which contain a sequence of testing commands. Multiple test cases can be grouped into a test suite so that they can be executed together. Each command usually includes a command name, a target element on the webpage, and an optional value.

Selenium IDE: https://www.selenium.dev/downloads

2.2 Automation tests

2.1.1 Tested websites

The website **https://www.mainguyen.vn/** is a Vietnamese e-commerce platform that focuses on selling technology devices and digital products. The website offers a variety of items such as smartphones, cameras, audio equipment, smart devices, and other electronic accessories.

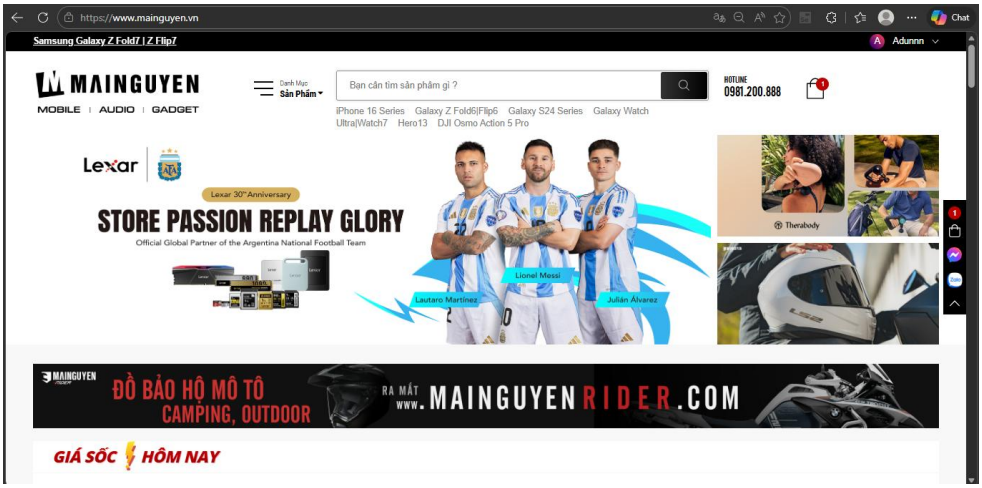

Figure 2.1 website https://www.mainguyen.vn/

The website provides several functions that support the online shopping experience. Users can browse products by category, view product listings, and open detailed product pages that display images, specifications, descriptions, and prices. This information helps customers understand the products before making a purchase.

The website also supports online purchasing, allowing users to add items to the shopping cart, manage selected products, and proceed with the buying process. Other features such as promotional sections, navigation menus, and keyword search help users easily find products.

Because the website includes many interactive elements such as menus, product pages, and shopping cart functions, mainguyen.vn is suitable for demonstrating automation testing using Selenium IDE. Testers can automate actions like browsing categories, opening product pages, and verifying website functionality to simulate real user behavior.

2.1.2 Test cases summary

TC ID	TC Description	Author	Marks
01	Verify login function	Trần Ngọc Ánh Dương	0.1
02	Verify logout function	Trần Ngọc Ánh Dương	0.1
03	Verify cart deletion function	Ao Thành Nhân	0.15
04	Verify total payment amount in the Shopping Cart	Ao Thành Nhân	0.15
05	Verify the quantity of products added to the cart	Ao Thành Nhân	0.15
06	Verify Product Attribute Selection	Trần Ngọc Ánh Dương	0.15
07	Verify Sorting by Product price	Ao Thành Nhân	0.2
08	Verify Product filter by Price Range	Ao Thành Nhân	0.15
09	Verify add to Wishlist functionality	Ao Thành Nhân	0.1
10	Verify product detail matches product card	Trần Ngọc Ánh Dương	0.15

11	Verify recently viewed product is displayed correctly	Ao Thành Nhân	0.1
12	Verify Buy Now redirects to Checkout with correct product	Trần Ngọc Ánh Dương	0.1
13	Verify only one product is added when clicking “Mua ngay”	Trần Ngọc Ánh Dương	0.1
14	Verify promotion code list display	Trần Ngọc Ánh Dương	0.1
15	Verify product search suggestion dropdown displays correctly	Trần Ngọc Ánh Dương	0.15
16	Verify product price is displayed within the selected price range	Trần Ngọc Ánh Dương	0.15
17	Verify selected color is correctly displayed on the product detail page	Trần Ngọc Ánh Dương	0.1
18	Update personal information	Trần Ngọc Ánh Dương	0.1
19	Verify search product that doesn’t exist	Ao Thành Nhân	0.1
20	Verify product that have comments on it	Ao Thành nhân	0.1
21	Verify percent discount of the product which has it is correct	Ao Thành Nhân	0.1
22	Verify number of product displayed on the page matched with the real number of products showed	Ao Thành Nhân	0.15
23	Verify search history deletion function	Ao Thành Nhân	0.15
24	Verify history search	Trần Ngọc Ánh Dương	0.15
25	Verify click on new page	Ao Thành Nhân	0.1

26	Verify order history	Trần Ngọc Ánh Dương	0.15
27	Verify cart icon navigation	Trần Ngọc Ánh Dương	0.05
28	Verify "Back to Top" button	Trần Ngọc Ánh Dương	0.05
29	Verify Social Media Links	Trần Ngọc Ánh Dương	0.05
30	Verify click on Logo will lead to the home page	Ao Thành Nhân	0.1

Table 2.1 Test cases summary

2.1.3 Test cases statistics

No.	Name	Quantity	Marks
1	Trần Ngọc Ánh Dương	16	1,75
2	Ao Thành Nhân	14	1,8

Table 2.2 Test cases statistics

2.1.4 Detailed test cases

TC ID	**Test Steps**	**Expected Result**
01	Verify product attribute selection (Apple option) Pre-con: User must have the account already 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on Đăng Nhập 3. Phone Number: 0768107838	The page displayed the name of the user and their avatar and no longer display the Login and Register button

	Password: TomTh0m@s 4. Click on Login button 5. Verify login successfully	
02	Verify logout function Pre-con: User must have their account and login already 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on their user name 3. Click on "Exit" 4. Verify the homepage after logout successful	The homepage will display "Register" and "Login" button
03	Verify Cart Deletion Functionality 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click Shopping Cart 3. Remove all products from the cart (if any exist) 4. Verify the cart status after removal	The message "Chưa có sản phẩm nào trong giỏ hàng" is displayed.
04	Verify the total amount in the Shopping Cart Pre-con: Clear the Cart	The total amount is correctly calculated based on the sum of product price added to the cart.

	1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Select menu Tai nghe 3. Select Monster brand 4. Add all available products to the shopping cart 5. Verify the total amount displayed in the cart	
05	Verify the quantity of products added to the cart Pre-con: Clear the Cart 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click “Máy tính bảng” category 3. Click “Samsung” brand 4. Click for each product and choose the first color option 5. Click “Thêm vào giỏ hàng” 6. Repeat steps 4 and 5 to add all the available products to cart 7. Verify cart quantity displayed in the home page	The cart quantity icon increases correctly according to the number of successfully added products.
06	Verify Product Attribute Selection 1. Navigation to: MaiNguyen.vn	The product names in the search results contain the keyword “Apple”.

	- Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Enter the "Tai Nghe" on the homepage. 3. Click "Apple" filter. 4. Verify all the displayed products	
07	Verify Sorting by Product price 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on "Mobile Phone" category 3. Click on "Low price" 4. Verify the product price sorting in the screen	The page will show the product price sorting from low to high
08	Verify Product filter by Price Range 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on "Phụ kiện/Đi tour 3. Click on Price range "10tr - 15tr" 4. Verify all the product prices displayed	All the product displayed have price inside the range
09	Verify add to Wishlist functionality	- The number of products added to the Wishlist is correct.

	Pre-con:User must already login 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Go to “Phụ kiện/Gimbal/ Phụ kiện vlog” 3. Select “Insta360” brand 4. Click add to wishlist for all products 5. Verify the number of Products added to Wishlist 6. Delete all products from the Wishlist	- The message “Danh sách trống” is displayed in the Wishlist
10	Verify product detail matches product card 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Enter keyword "iphone11" into the search bar. 3. Click the Search button. 4. Select the first product from the search results. 5. Store the product name and price displayed on the product card. 6. Click on the selected product to open the product detail page. 7. Store the product name and	The product name and price on the detail page matches the product name and price on the product card.

	price displayed on the product detail page. 8. Compare the product name and price between the product card and the product detail page.	
11	Verify Product Attribute Selection 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Search for the keyword "iPhone". 3. Click on any product from the product list. 4. Store the product title of the selected product. 5. Click on the user name (header). 6. Select "Thông tin cá nhân". 7. Navigate to "Sản phẩm đã xem" section. 8. Get the title of the first product in the "Recently Viewed" list. 9. Verify that the title matches the stored product title from step	The "Recently Viewed" section in the user profile displays the correct product that was just clicked The product title in "Recently Viewed" matches exactly with the title of the previously selected product
12	Verify Buy Now redirects to Checkout with correct product 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên |	The product name on the checkout page matches the name on the product detail page exactly.

	mainguyen.vn 2. Click on the "Accessories" category. 3. Click on any product. 4. Store the product name from the product detail page. 5. Click the "Buy Now" button. 6. Verify the product name displayed on the checkout page.	
13	Verify only one product is added when clicking “Mua ngay” 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on the "Accessories" category. 3. Click on any product. 4. Click the "Buy Now" button. 5. Locate the quantity input field (between “-” and “+” buttons). 6. Get the quantity value displayed. 7. Verify the quantity value.	Quantity of the product is exactly 1.
14	Verify promotion code list display 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn	The list of discount codes is displayed correctly: • If no code is found, display "Empty list" • If a code is found, display a list of valid codes.

	2. Click on the "Accessories" category. 3. Click on any product (ví dụ sản phẩm thứ 6). 4. Click “Dùng mã giảm giá”. 5. Check the promotion code list content.	
15	Verify product search suggestion dropdown displays correctly 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on the search bar at the top of the page.. 3. Type the keyword "ip". 4. The dropdown displays a list of relevant product suggestions related to "ip".	The dropdown displays a list of relevant product suggestions related to "ip".
16	Verify product price is displayed within the selected price range 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Select menu “Điện thoại”. 3. Locate the Price Filter section. 4. Enter the minimum price in the “From” field. 5. Enter the minimum price in	All displayed product prices fall within the selected price range.

	the “From” field. 6. Click the “Áp dụng” button. 7. Verify that the price of each product card is greater than or equal to the minimum price and less than or equal to the maximum price entered.	
17	Verify selected color is correctly displayed on the product detail page 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Select menu “Điện thoại”. 3. Locate the color filter section on the page. 4. Click the "Pink" color in the color filter. 5. Verify the color displayed on the product detail page.	Verify that the selected color on the detail page matches the color chosen in the filter.
18	Update Personal Information 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click the user name displayed on the header. 3. Select “Thông tin cá nhân” 4. Update the name by typing “Adunnn” into the name field. 5. Click the “Lưu thay đổi” button to submit the changes.	Verify that the user name displayed on the header is updated to “Adunn”.

	6. Verify that the user name displayed on the header is updated to “Adunnn.”	
19	Verify search product that doesn’t exist 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on Search’s bar 3. Search for product “Realme” which doesn’t exist. 4. Click search 5. Verify which is the screen displayed	The screen displayed “Kết quả tìm kiếm Realme - 0 sản phẩm’
20	Verify product that have comments on it 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on Search’s bar 3. Search for product “dán cường lực galaxy s24 ultra” or navigation to this url in case you can’t find it (https://www.mainguyen.vn/dan-cuong-luc-galaxy-s24-ultra-mipow-kingbull-hd-premium-silk-p57354) inside MaiNguyen website	The number of comments displayed is matching with the comments that products have

	4. Click Search 5. Click on products which have comments on it Verify number of comments displayed on the page	
21	Verify percent discount of the product which has it is correct 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Đông Mai Nguyên | mainguyen.vn 2. Click search then type "Galaxy Tab S8 Ultra 128GB - 8GB" 3. Click search 4. Verify the percent discount of that product using the formula 5. Compare with the percent discount displayed on the page	The percent discount displayed on the page is correct
22	Verify number of products displayed on the page matched with the real number of products showed 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Đông Mai Nguyên | mainguyen.vn 2. Click on "Smarthome - Nhà thông minh" category	The number products showed in the page is correct with the real number of products

	3. Verify number of products displayed on the page compare with the real number of products showed	
23	Verify search history deletion function Pre-con: User must have search history 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on Search bar 3. Click delete for each Search history until it's clear 4. Verify the search history after the removal	The search history is empty
2	Verify search history 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on search bar 3. Type "Samsung" then click Search 4. Back to main page 5. Type "Iphone" the click Search 6. Back to main page again	The history search showed "Samsung" search and "Iphone" search and sort Samsung under Iphone keyword

	7. Click on Search bar again 8. Verify that the history search should include previous search and sorted them	
25	Verify click on new page 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on "Điện thoại" category 3. Scroll down then click on page 2 4. Verify that the product list has changed	The page shows the new product list on page 2
26	Verify order history 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click the user name displayed on the header. 3. Select "Thông tin cá nhân" 4. Select "Đơn hàng" from the account menu. 5. Verify that the Order History page is displayed.	The system correctly displays the list of user orders with full order information.
27	Verify Cart Icon Navigation 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên |	Verify cart page is displayed

	mainguyen.vn 2. Click cart icon	
28	Verify "Back to Top" button 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Scroll down to the bottom of the page. 3. Click the "Up Arrow" button	The page smoothly scrolls back to the top
29	1. Verify Social Media Links 2. 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 3. 2. Scroll down to Footer. 4. 3. Click Facebook icon. 5. Verify the link attribute of the icon.	The href attribute of the Facebook icon must be https://www.facebook.com/mainguyencenter
30	Verify click on Logo will lead to the home page 1. Navigation to: MaiNguyen.vn - Công Ty TNHH Công Nghệ Di Động Mai Nguyên | mainguyen.vn 2. Click on “Đồng hồ thông minh” category 3. Then click on the logo “MAINGUYEN” on top left of	The home page is showed after click on the logo

	the page 4. Verify that after click on the logo it will return to the home page	

Table 2.3 Detailed test cases

CHAPTER 3. CONCLUSION

In this chapter, the fundamental concepts of automated web testing using Selenium and TestCafe were presented. These two frameworks are widely used tools for performing automated testing on web applications, helping developers and testers improve testing efficiency, reduce manual effort, and ensure software quality.

Selenium is one of the most popular and established automation testing frameworks. It provides powerful tools such as Selenium WebDriver that allow testers to control web browsers programmatically and simulate real user interactions. Because of its flexibility and large community support, Selenium has become a standard solution for automated web testing in many software development projects.

TestCafe, on the other hand, is a modern automation testing framework designed to simplify the testing process. Built on Node.js, TestCafe allows testers to write tests using JavaScript or TypeScript without the need for additional browser drivers or complex configurations. One of its advantages is the automatic waiting mechanism, which improves test stability by handling asynchronous actions automatically. TestCafe also supports cross-browser testing and provides a simple syntax for writing test scripts.

Through the introduction of these two frameworks, this chapter highlights their roles in modern software testing. Selenium offers flexibility and extensive ecosystem support, while TestCafe focuses on simplicity and ease of use. Both tools provide efficient solutions for automating web application testing and improving the reliability of software systems.

These theoretical foundations will serve as the basis for the implementation and practical testing demonstrations presented in the following chapters.

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